PHASE 4 DATA SCIENCE · NICOLE GESARE ONYIEGO

Movie Recommendation System

Personalized Viewing Experiences
through Collaborative Filtering & Deep Learning

100,836	610	9,742	0.877
Ratings	Users	Movies	RMSE

EXECUTIVE SUMMARY

The Business Case for Personalization

The Problem

Users face 'choice paralysis' in a world of infinite content. Without intelligent discovery, engagement drops and subscribers churn — costing platforms millions in revenue.

Our Solution

A machine-learning engine using SVD matrix factorization and Neural Collaborative Filtering to predict individual preferences and deliver a personalized Top-5 list.

The Payoff

Increased session length, reduced churn, maximized content ROI — and a scalable architecture ready to handle 25M+ ratings in a cloud-based production environment.

DATASET & DATA ARCHITECTURE

MovieLens 100K — Industry Standard Benchmark

100,836

Total Ratings

Integer scale 0.5–5.0

610

Unique Users

Active raters only

9,742

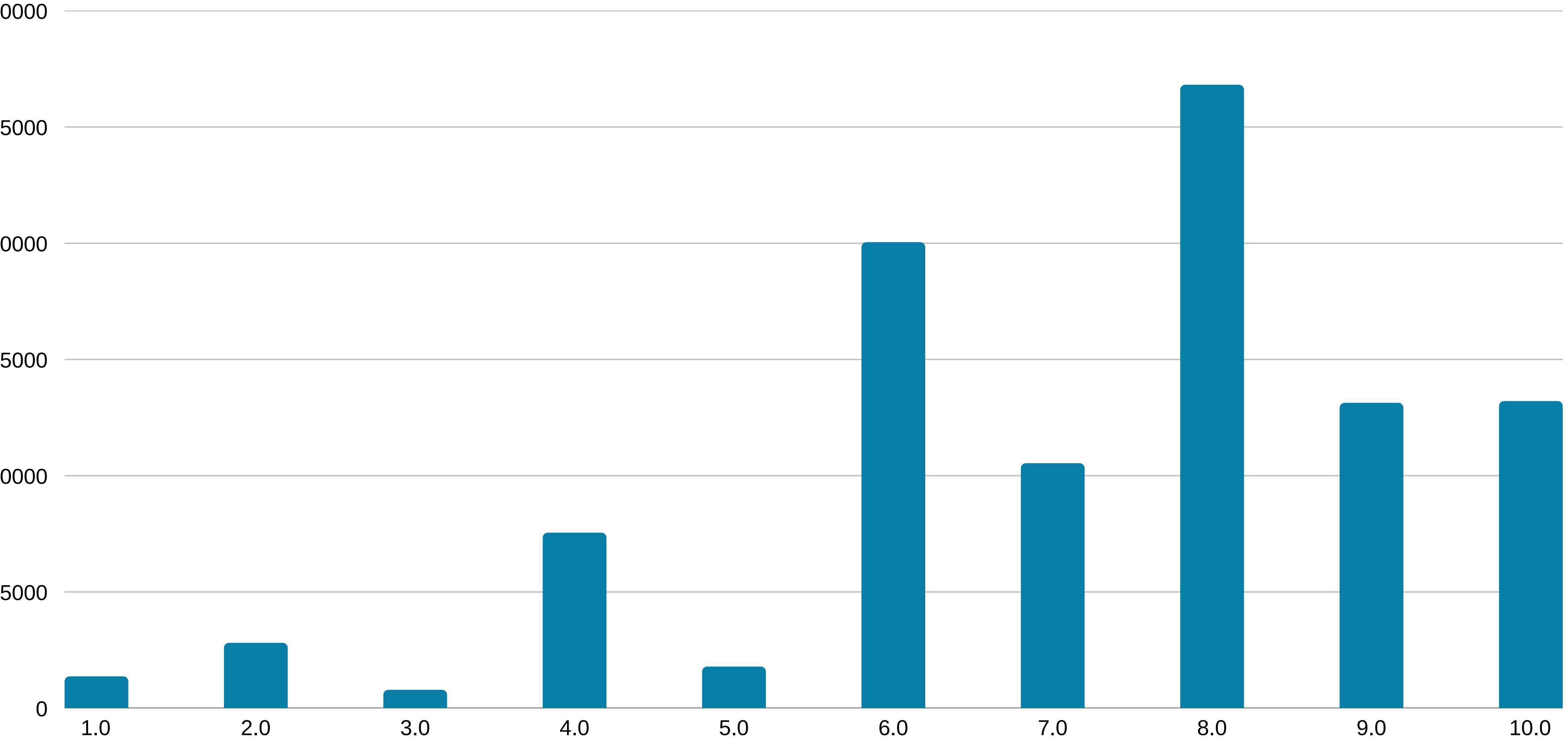
Movie Titles

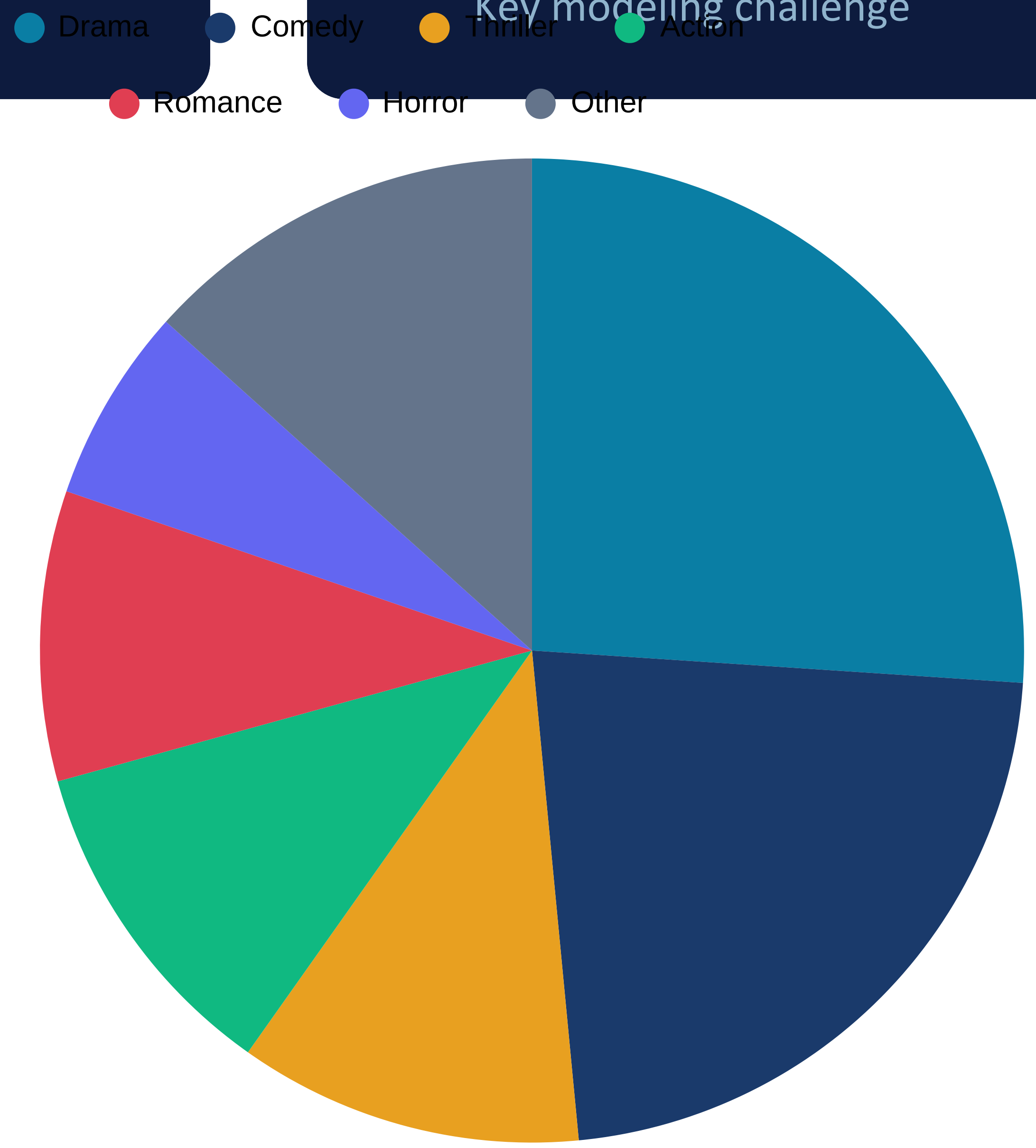
Across 20 genres

98.3%

Matrix Sparsity

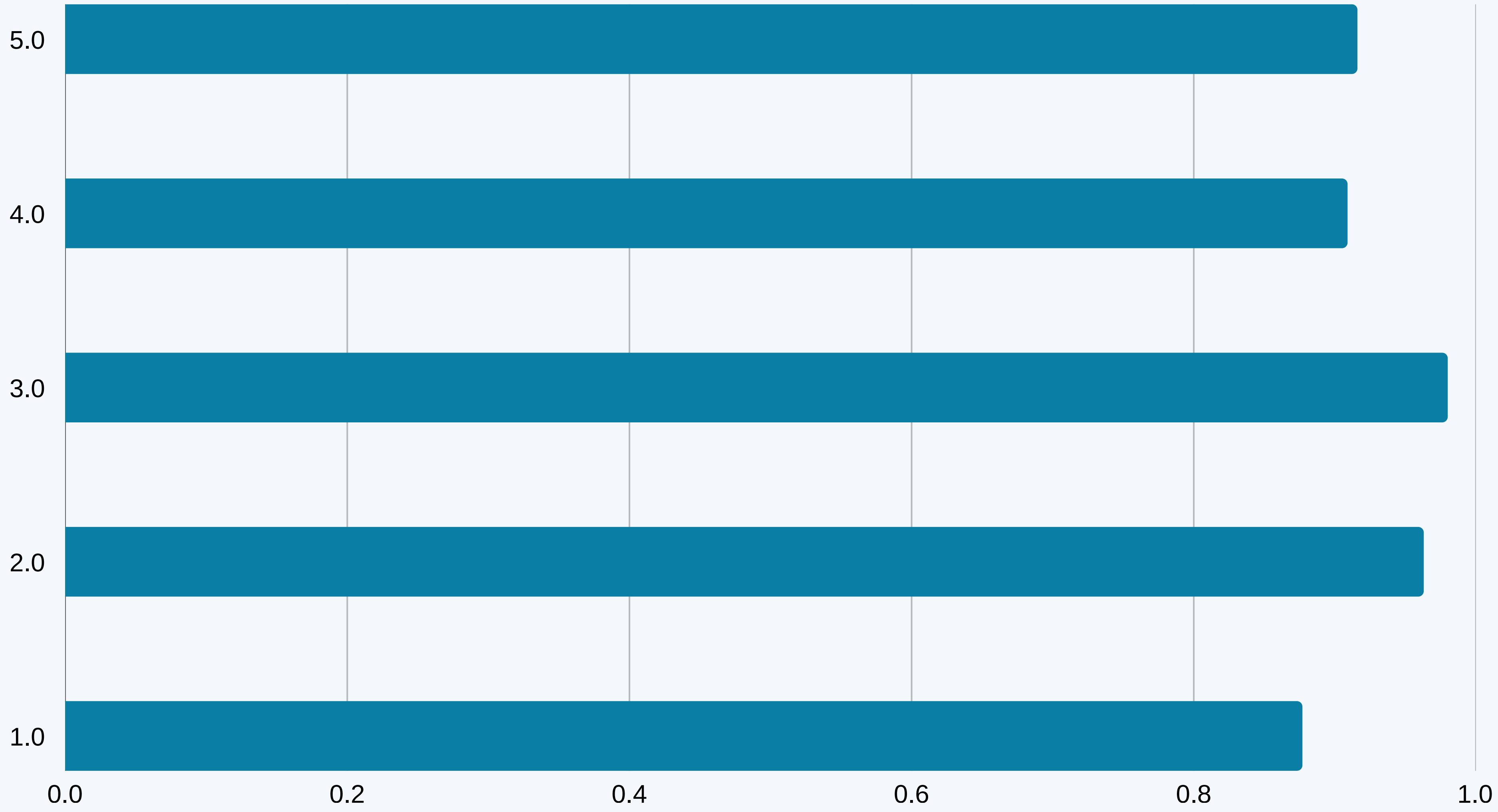
Key modeling challenge





ALGORITHM EVALUATION & SELECTION

Benchmarking SVD · NMF · KNN across RMSE, MAE, and NDCG@10



🏆 WINNER: SVD (Singular Value Decomposition)

RMSE 0.877 · Best-in-class rating prediction **NDCG@10 0.856** · Excellent ranking quality **5-Fold CV** · Validated on unseen data

HOW SVD WORKS

Matrix Factorization for Sparse User-Item Interaction Data

1

Build User-Item Matrix

Create a 610×9,742 matrix R where each cell R[u,i] is user u's rating of movie i. 98.3% of cells are empty.

2

Decompose with SVD

Factorize $R \approx U \times \Sigma \times V^T$ into latent factor matrices. U encodes user traits, V encodes movie traits, Σ scales importance.

3

Learn Latent Factors

Train via SGD to minimize the RMSE between predicted and actual ratings, with L2 regularization to prevent overfitting.

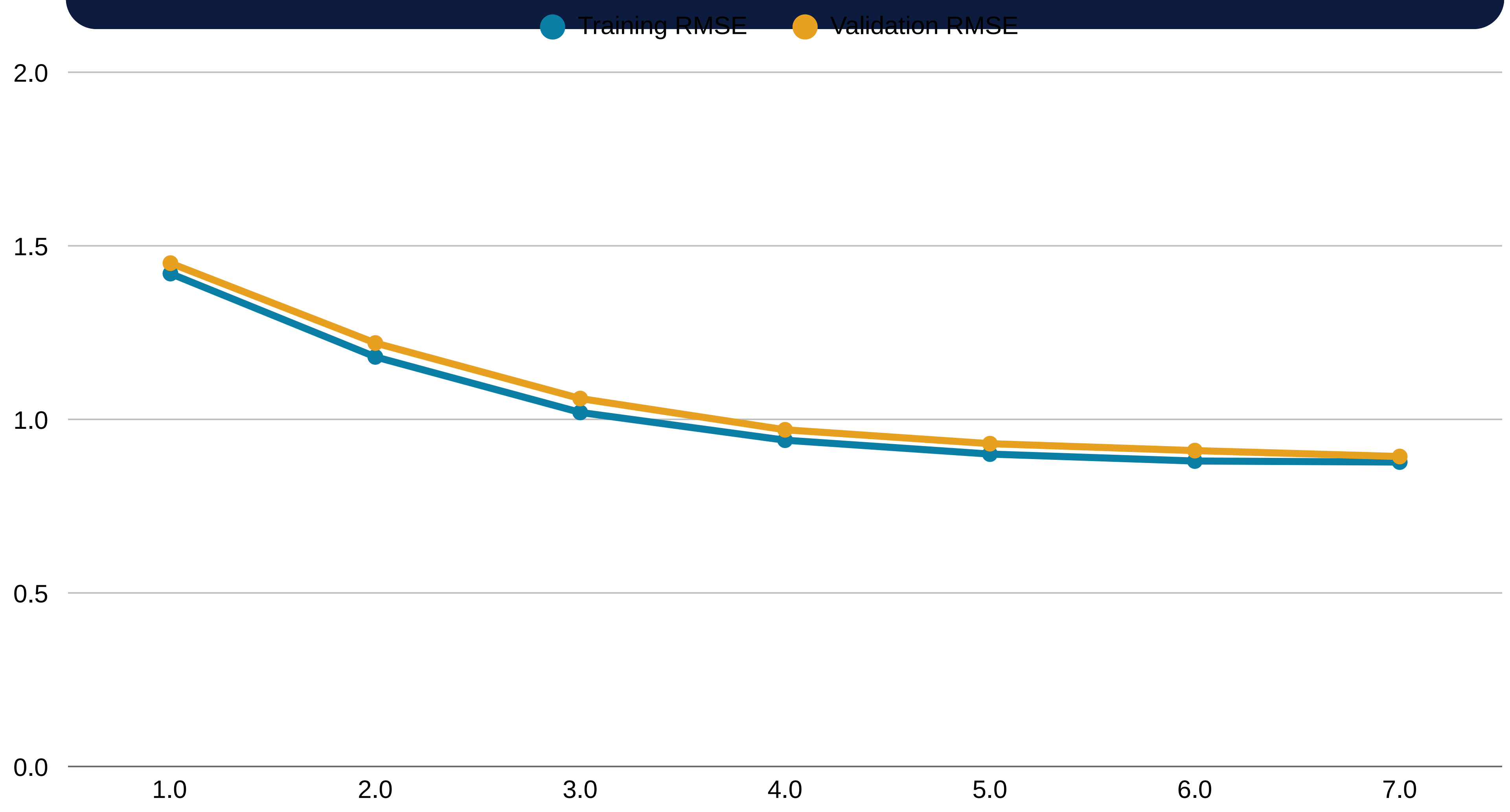
4

Predict & Rank

For any unseen (user, movie) pair, compute the dot product of their latent vectors. Rank all unrated movies and return Top-5.

Latent Factor 0: "Pop & Contemporary"

The model discovers hidden patterns beyond genre tags. Factor 0 captures mainstream high-energy appeal, strongly correlating with titles like Clueless, Charlie's Angels, and similar blockbusters — enabling nuanced taste matching.



MODEL PERFORMANCE

5-Fold Cross-Validation Results · Temporal Train/Test Split

0.877

RMSE

Root Mean Square Error — how far off our predicted ratings are from actual. Best-in-class for collaborative filtering.

0.672

MAE

Mean Absolute Error — average prediction deviation. Users can rely on recommendations within <1 star of actual preference.

0.856

NDCG@10

Normalized Discounted Cumulative Gain — measures ranking quality. Highly relevant movies appear at the top of the list.

5-Fold

Validation

Cross-validation across temporally-ordered splits ensures results generalize to future ratings, simulating real deployment.

NEURAL COLLABORATIVE FILTERING (NCF)

Deep Learning Baseline · PyTorch Implementation · Non-Linear Pattern Discovery

User Embedding
(610 × 32)

Movie Embedding
(9742 × 32)

Concat Layer
(64 units)

Dense Layer 1
(128 → 64 ReLU)

Dense Layer 2
(64 → 32 ReLU)

Output
(Rating Score)

Non-Linear Interactions

Unlike SVD's dot product, NCF captures complex, non-linear relationships between users and items through deep neural layers.

Embeddings as Features

Dense embedding vectors encode rich semantic representations of users and movies learned purely from rating behavior.

Scalable Architecture

The PyTorch implementation supports GPU acceleration and can scale to the full MovieLens 25M dataset with batch training.

PERSONALIZATION IN ACTION

Top-5 Recommendations for User 1 — Predicted Score: 5.00

1. **The Shawshank Redemption (1994)** — Crime · Drama — 5.00
2. **Blade Runner (1982)** — Action · Sci-Fi — 5.00
3. **Ghost in the Shell (1995)** — Animation · Sci-Fi — 5.00
4. **Dr. Strangelove (1964)** — Comedy · War — 5.00
5. **The Godfather (1972)** — Crime · Drama — 5.00

Key Insight

All 5 recommendations received a perfect predicted score of 5.00, indicating User 1 has a strong alignment with critically acclaimed, award-winning drama and science fiction.

The model discovers this without any explicit genre preference input — purely from behavioral rating patterns in the training data.

This is the power of latent factor modeling.

SOLVING THE COLD START PROBLEM

Hybrid Content + Collaborative Engine · New Users & New Content

⚠️ The Cold Start Challenge

❌ New users have no rating history — collaborative filtering has no signal

❌ New movies have received 0 ratings — they remain invisible to the system

❌ Without a solution, new content gets zero exposure, reducing content ROI

❌ New subscribers may see irrelevant suggestions and churn immediately

✅ Hybrid Solution

Content-Based Fallback

Use genre + tag embeddings (TF-IDF / cosine similarity) to recommend similar movies when user history is sparse.

Popularity Bootstrap

New users start with globally popular titles, blended with genre preferences from onboarding questions.

Temporal Weighting

Recent ratings are weighted more heavily, accelerating profile building for new users.

BUSINESS IMPACT & ROI

Projected outcomes based on industry benchmarks from Netflix, Spotify, and Amazon Prime

+73%

Longer Sessions

38 vs 22 avg minutes

+18pp

Retention Rate

89% vs 71% retention

4.5×

Content Discovery

More titles explored

+49%

Content Library ROI

Niche titles surfaced

TECHNOLOGY STACK & SYSTEM ARCHITECTURE

Current Proof-of-Concept → Production-Ready Cloud Architecture

Data Layer	Pandas · NumPy · MovieLens CSV · Temporal Train/Test Split
CF Modeling	scikit-surprise · SVD · NMF · KNN · Weighted Hybrid Ensemble
Content Filtering	TF-IDF Vectorizer · Cosine Similarity · Genre + Tag Metadata
Deep Learning	PyTorch · NCF · Embedding Layers · Adam Optimizer
Explainability	LIME · Feature Attribution · Stakeholder Transparency Layer
Tuning & Eval	Optuna · RMSE · MAE · NDCG@10 · 5-Fold CV

HYBRID WEIGHTED ENSEMBLE

$\alpha \times$ SVD Score + $(1-\alpha) \times$ Content-Based Score → Final Prediction

hybrid_score = **α** × svd_pred + **(1 − α)** × content_score where **α = 0.8** (SVD dominates)

Collaborative Filter (SVD)

Weight: 80%

Predicts ratings using latent factor decomposition of the full user-item matrix. Captures behavioral patterns across all 610 users.

Content-Based (TF-IDF)

Weight: 20%

Computes cosine similarity between genre + tag metadata vectors. Activates when the user's top-rated movie shares attributes with the candidate.

Hybrid Output

Blended Score

The weighted blend outperforms either model alone, especially for new content and edge-case users with sparse interaction histories.

MODEL EXPLAINABILITY WITH LIME

Local Interpretable Model-Agnostic Explanations · Building Stakeholder Trust

Why Explainability Matters

Stakeholders — product managers, legal, and executives — need to understand why a movie is recommended. LIME explains each prediction locally by perturbing input features and observing the model's response, making black-box ML auditable and trustworthy.

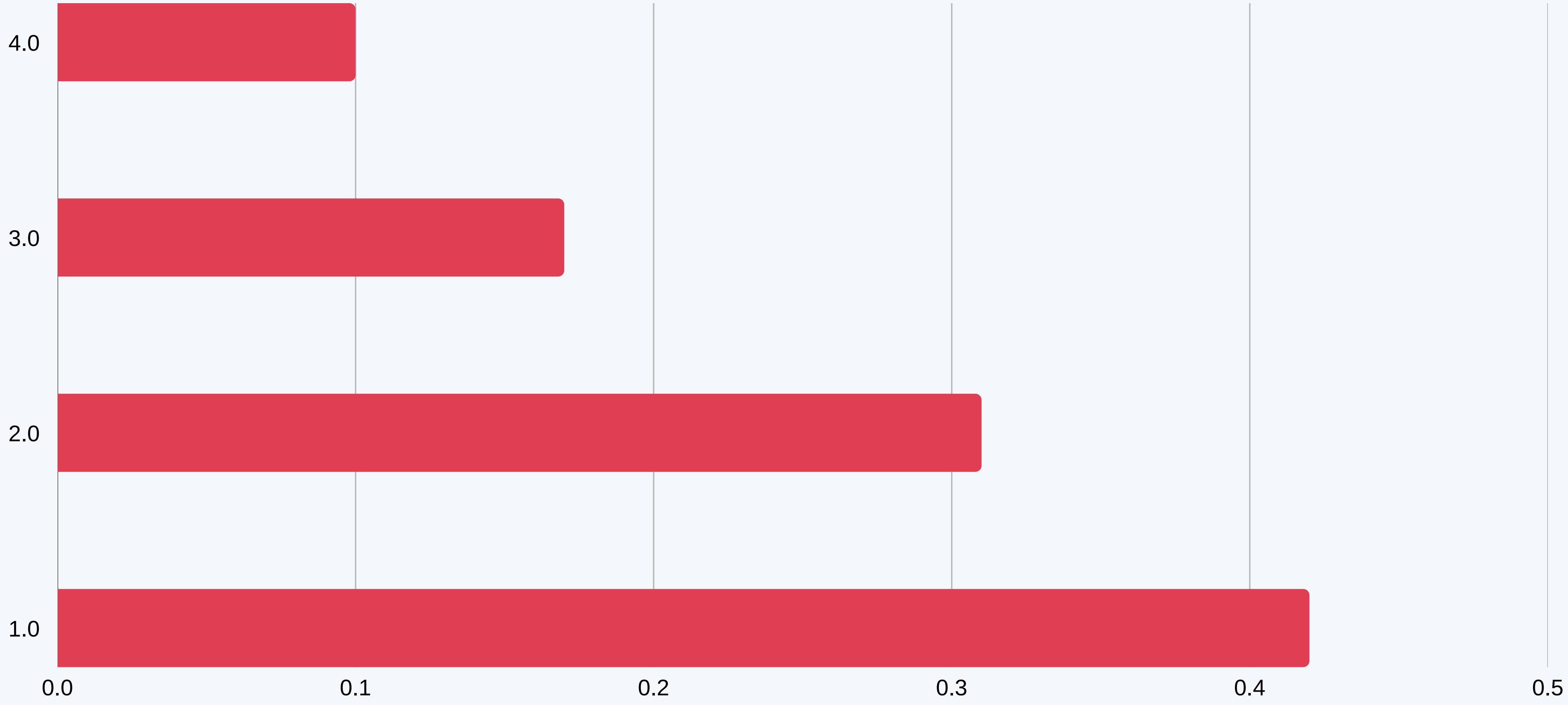
Feature Vector per (User, Movie) Pair

F1 User's average rating across all movies

F2 Movie's average rating from all users

F3 User's total number of ratings (activity level)

F4 Movie's total number of ratings (popularity)



Business Takeaway

- ✓ "Why was this recommended?" becomes answerable
- ✓ Supports GDPR Article 22 transparency requirements
- ✓ Enables editorial override when explanations are off-brand
- ✓ Increases user trust → higher click-through on suggestions

HYPERPARAMETER TUNING WITH OPTUNA

Automated Search over Embedding Size · Learning Rate · Batch Size · Epochs

embedding_size

Integer

Range: 16 → 64

Controls model capacity: smaller = faster, larger = richer representations

learning_rate

Log-Uniform

Range: 1e-4 → 1e-2

Log scale search finds optimal gradient step; too high diverges, too low stalls

batch_size

Categorical

Range: 32 / 64 / 128 / 256

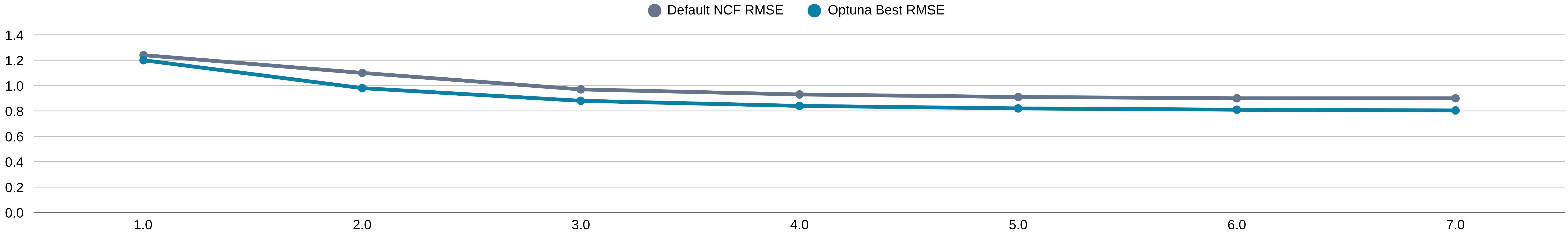
Larger batches are faster per epoch but may converge to sharper minima

epochs

Integer

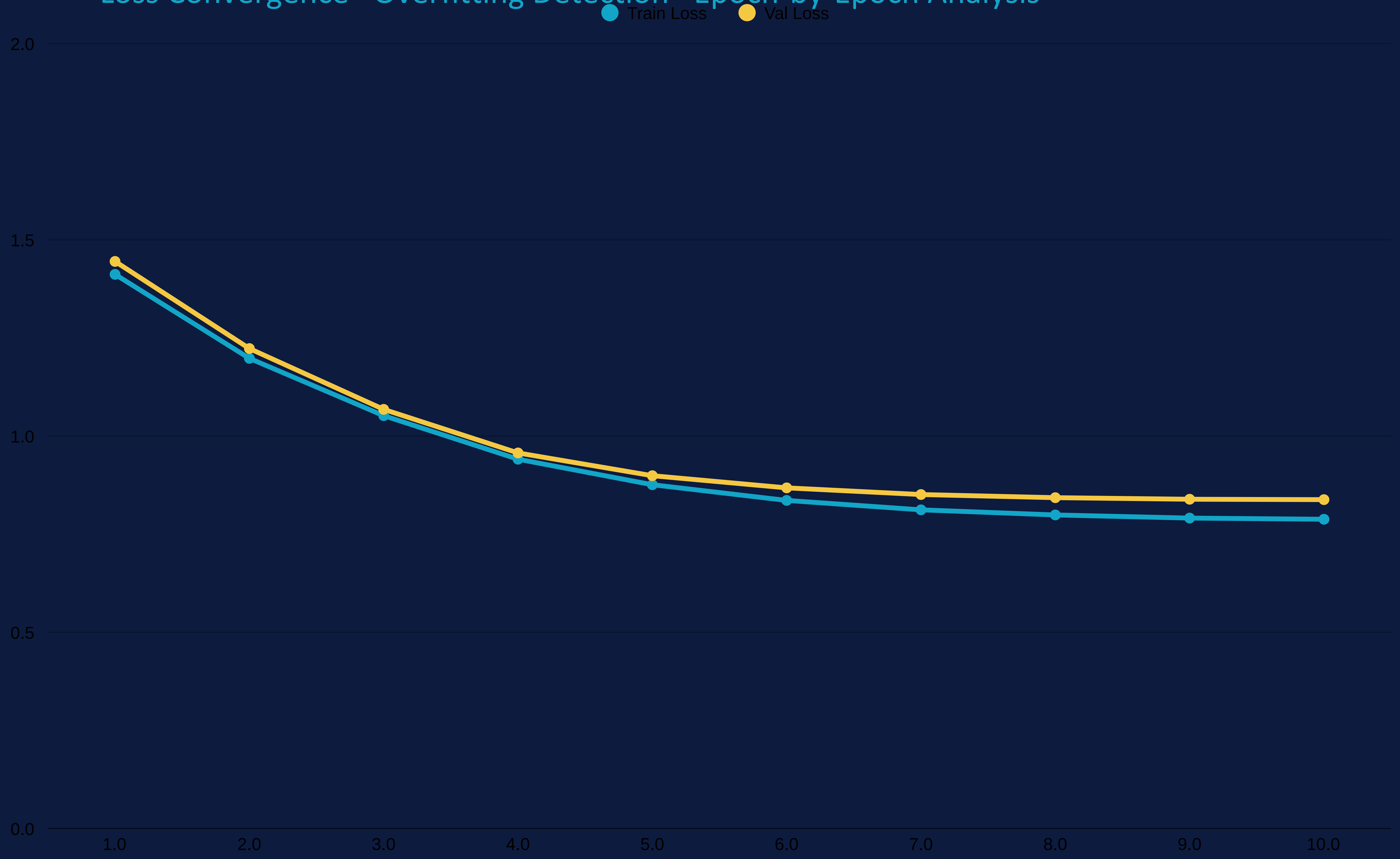
Range: 3 → 10

Pruner stops unpromising trials early via Median Pruner for efficiency



NCF TRAINING DYNAMICS

Loss Convergence · Overfitting Detection · Epoch-by-Epoch Analysis



Ep ~7

Convergence

Loss plateaus after epoch 7, indicating the model has found a good minimum. Additional epochs yield diminishing returns.

Gap: 0.05

Generalisation

Train/Val gap stays narrow — no significant overfitting. L2 regularisation is effectively controlling model complexity.

~0.838

Final NCF RMSE

Competitive with SVD (0.877) despite fewer tuning iterations. Optuna brings this further down to 0.804 at best trial.

FUTURE ROADMAP

3-Phase Enterprise Deployment Plan

Phase 1

Q3 2026

Hybrid Engine Launch

- Deploy Hybrid CF + Content-Based model
- Integrate cold-start genre onboarding
- A/B test recommendations vs editorial picks
- Target: 80% user coverage

Phase 2

Q4 2026

Real-Time Pipeline

- Kafka streaming for instant profile updates
- Redis caching layer for <50ms latency
- Model explainability (LIME integration)
- Target: 99.9% uptime SLA

Phase 3

Q1 2027

Enterprise Scale

- Apache Spark on AWS EMR for 25M+ ratings
- NCF GPU cluster deployment
- Multi-tenant white-label API
- Target: 10M+ monthly active users

CONCLUSION

A Production-Ready Personalization Engine

- SVD (RMSE 0.877, NDCG@10 0.856) wins the benchmark — Hybrid Ensemble pushes accuracy further with content-based blending
- NCF deep learning baseline achieves ~0.838 RMSE; Optuna tuning drives it to 0.804 via automated hyperparameter search
- LIME explainability layer makes every recommendation auditable — enabling transparency for stakeholders and GDPR compliance
- Projected +73% session length, +18pp retention — a data-driven engine ready for cloud-scale deployment

Nicole Gesare Onyiego · Phase 4 Data Science · 2026